

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/B_DEB_SAALKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 3663, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.0001,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505,   0.1, 0.001,    2);
Distrib (kap,         Uniform,                0,        1);
Distrib (kapA,        Uniform,                0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,   0.3);
Distrib (K,           Uniform,                0.00001,   2);
Distrib (dv,          TruncNormal_cv,        2,      0.5, 0,      10);
Distrib (wE,          LogUniform,          0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, RTOL, ATOL)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

Fm.1. pAm.1. r_pAm_pM.1. v.1. kap.1. kapA.1.

Result_mode	0.005037700	164.10600	0.76857000	0.20176400	0.92553100	0.52867600
2.5%	0.003176199	148.86200	0.64723275	0.15302000	0.88880545	0.45267915
97.5%	0.011142490	198.29600	0.87804045	0.22363700	0.95097800	0.58727200
Min	0.002292030	94.58760	0.60395500	0.12563400	0.50515300	0.18063900
Max	0.346466000	214.07300	0.94010000	0.25924200	0.96090700	0.73462200
Result_mean	0.006629571	171.03613	0.76704078	0.18766372	0.92047964	0.52351693
Result_sd	0.003378871	12.87301	0.05959266	0.01797114	0.01593519	0.03519446
Ehp.1. E_coc.1. rOM_ClxHorse.1. K.1. dv.1.						
Result_mode	96.46380	26.41610	0.0071932500	0.03389010	4.2342700	
2.5%	73.52230	21.39705	0.0046274175	0.02109030	3.1493300	
97.5%	135.95900	36.57040	0.0101511800	0.06897790	5.7689010	
Min	63.31420	18.10510	0.0000161998	0.01795560	0.5220400	
Max	159.58700	44.60300	0.0130906000	0.98193800	6.6706600	
Result_mean	101.77891	28.36575	0.0073408584	0.04033105	4.2587670	
Result_sd	16.23677	3.85009	0.0014061268	0.01441911	0.6581547	
wE.1. Sigma_W.1.						
Result_mode	1.015720e-09	0.12017400				
2.5%	1.726510e-09	0.10816100				
97.5%	1.044180e-05	0.13742700				
Min	1.015720e-09	0.09847980				
Max	1.172510e-05	5.72094000				
Result_mean	3.141676e-06	0.12246482				
Result_sd	3.707704e-06	0.05372067				

Number of observations, n = 277
 Number of parameters, k = 12
 Log-likelihood, LL = -17.30957
 BIC = 102.1074

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.08	1.26
E_coc.1.	1.04	1.13
Ehp.1.	1.19	1.53
Fm.1.	1.24	1.68
K.1.	1.21	1.59
kap.1.	1.20	1.64
kapA.1.	1.02	1.06
pAm.1.	1.07	1.18
r_pAm_pM.1.	1.16	1.47

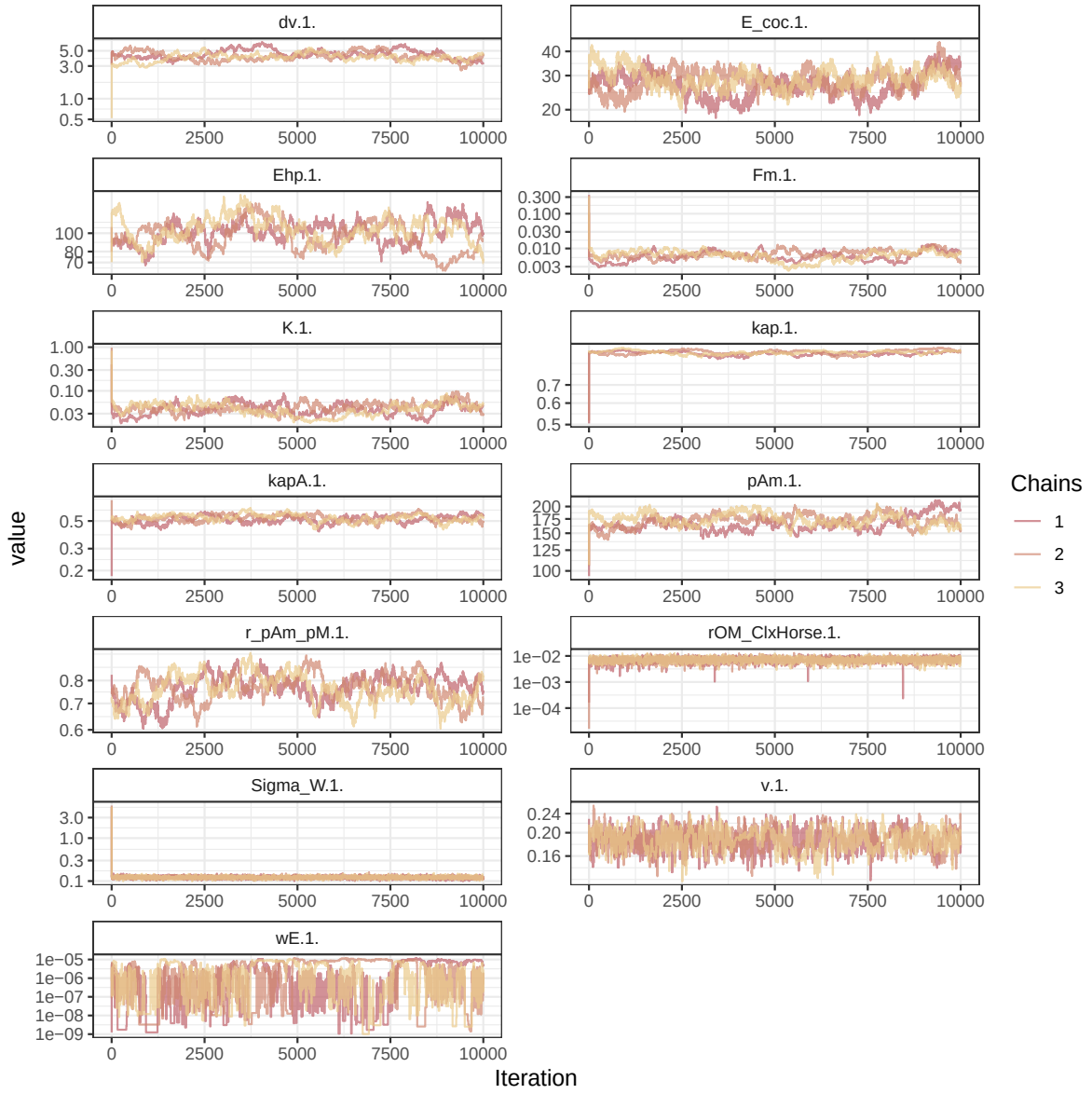


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

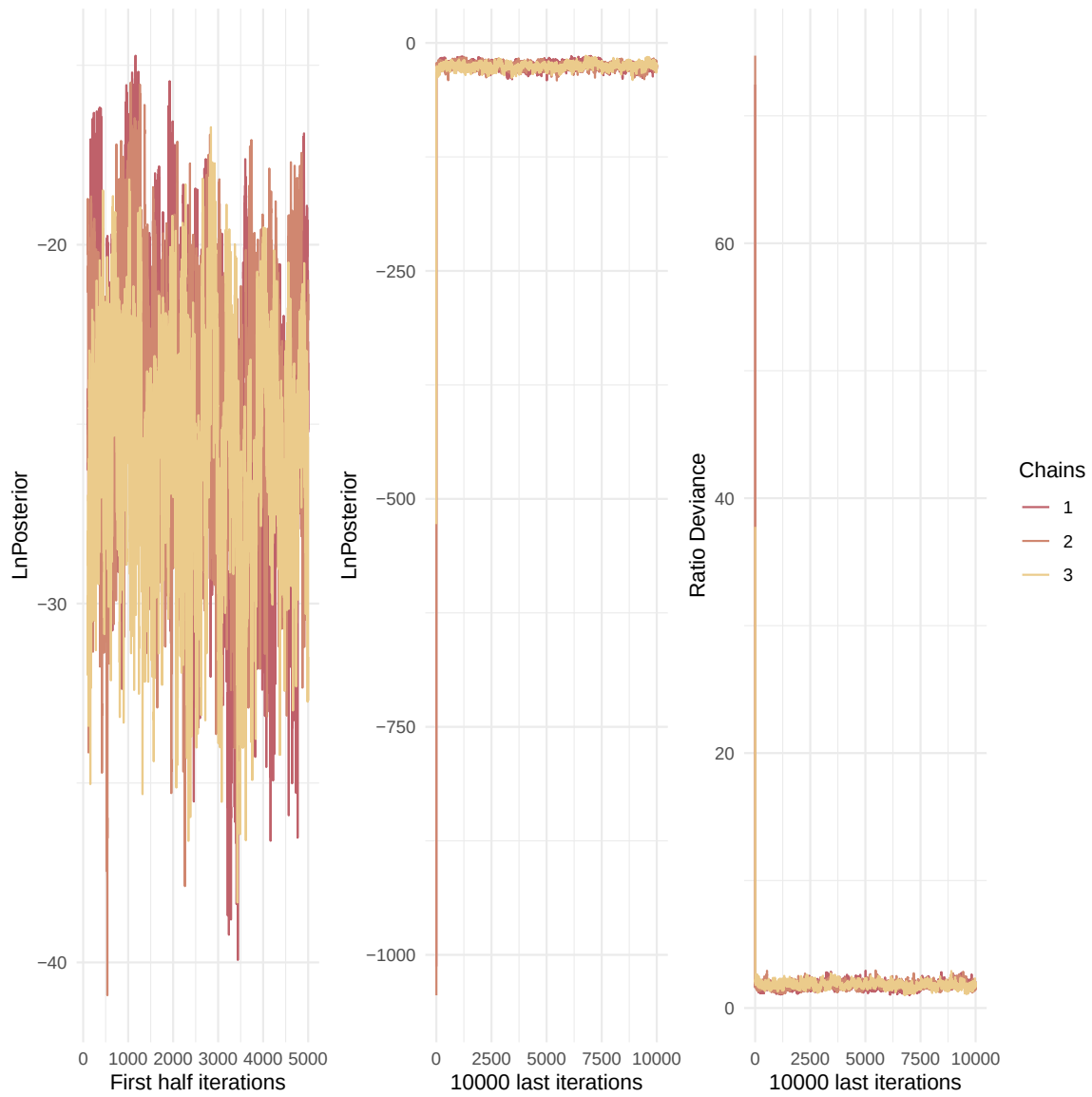


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

rOM_ClxHorse.1.	1.01	1.03
Sigma_W.1.	1.01	1.02
v.1.	1.01	1.04
wE.1.	1.26	1.77

Multivariate psrf

1.53

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

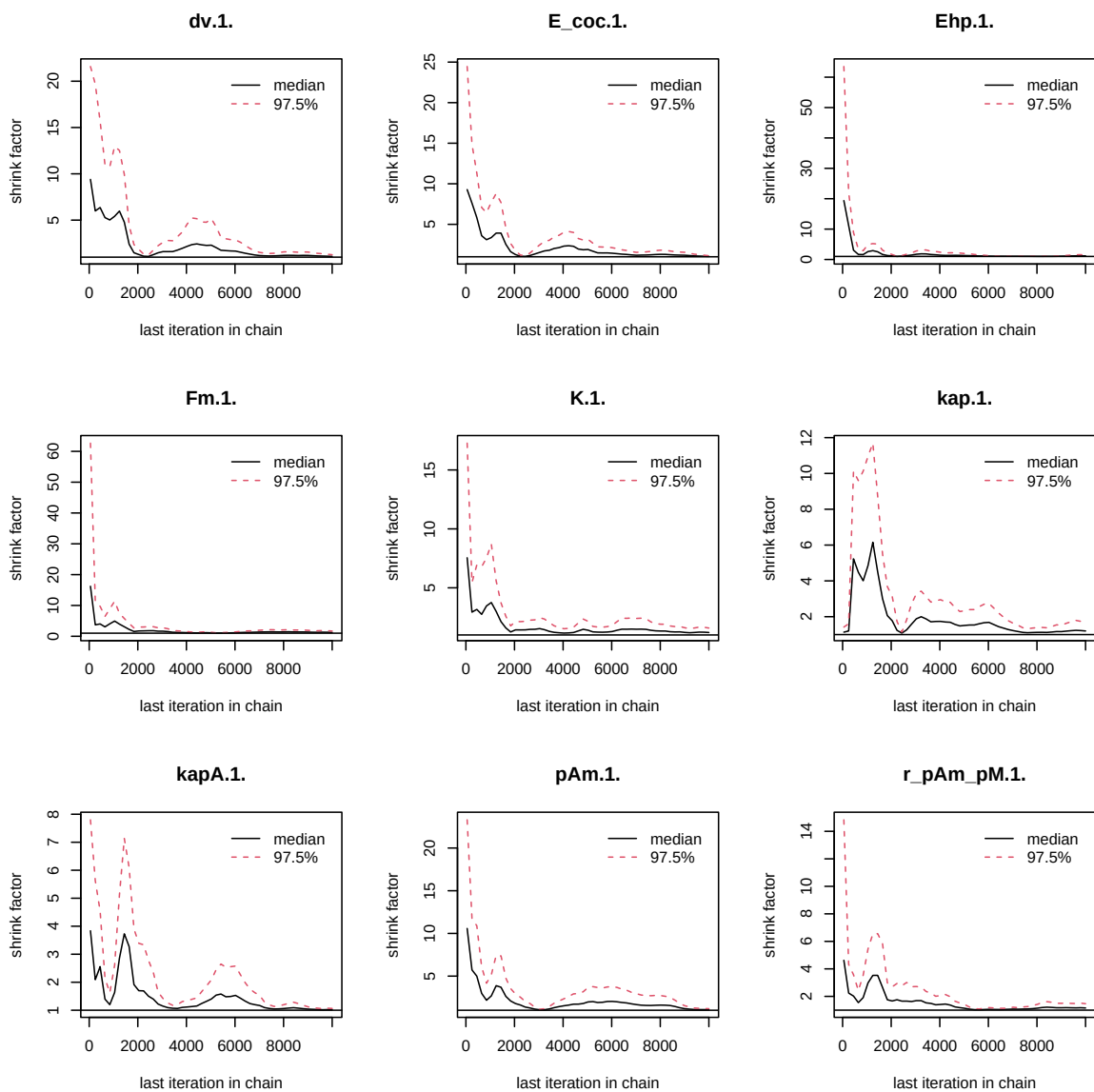


Figure 3: Chains convergence.

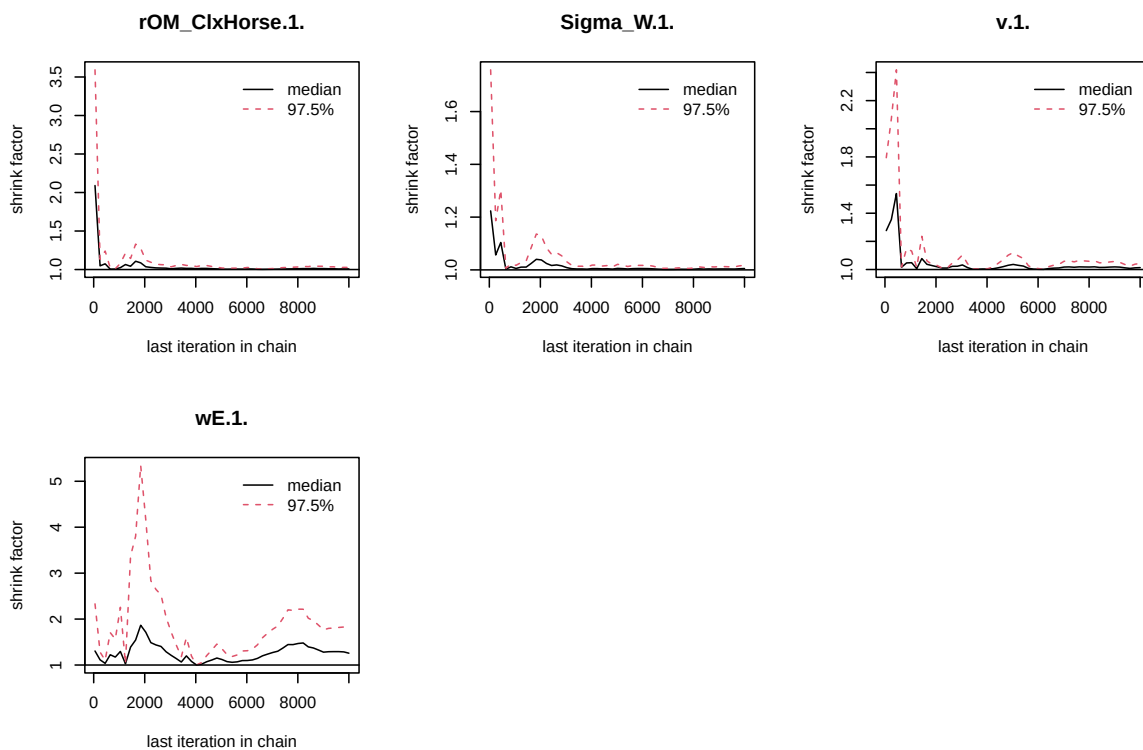


Figure 4: Chains convergence.

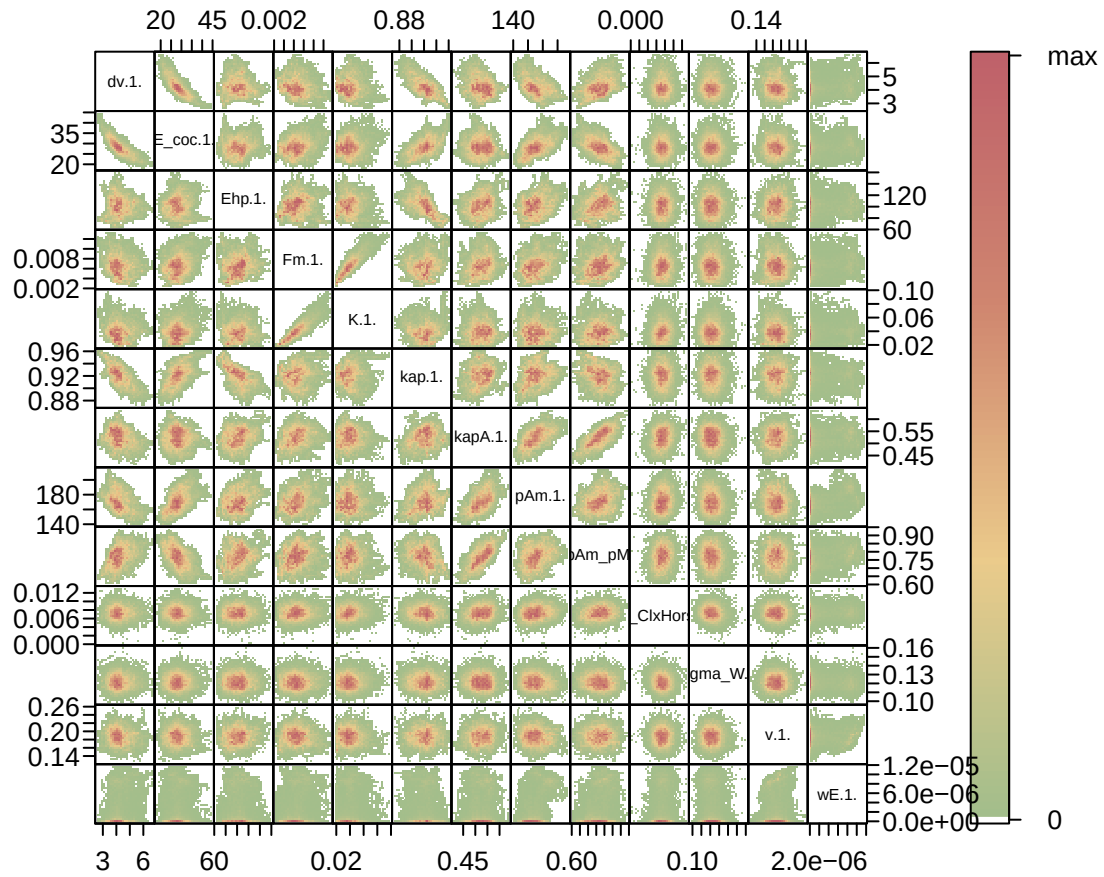


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",

```

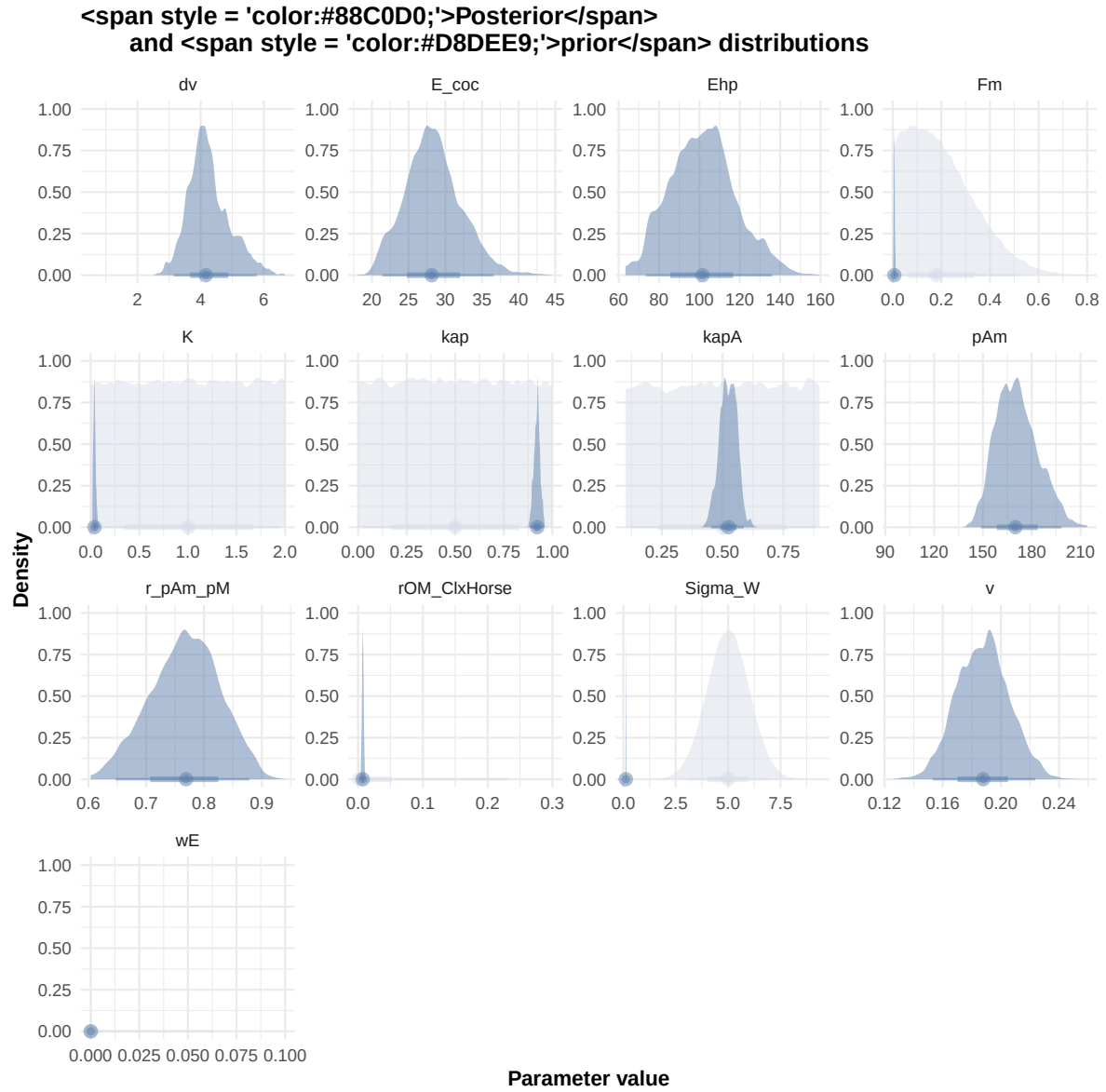


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

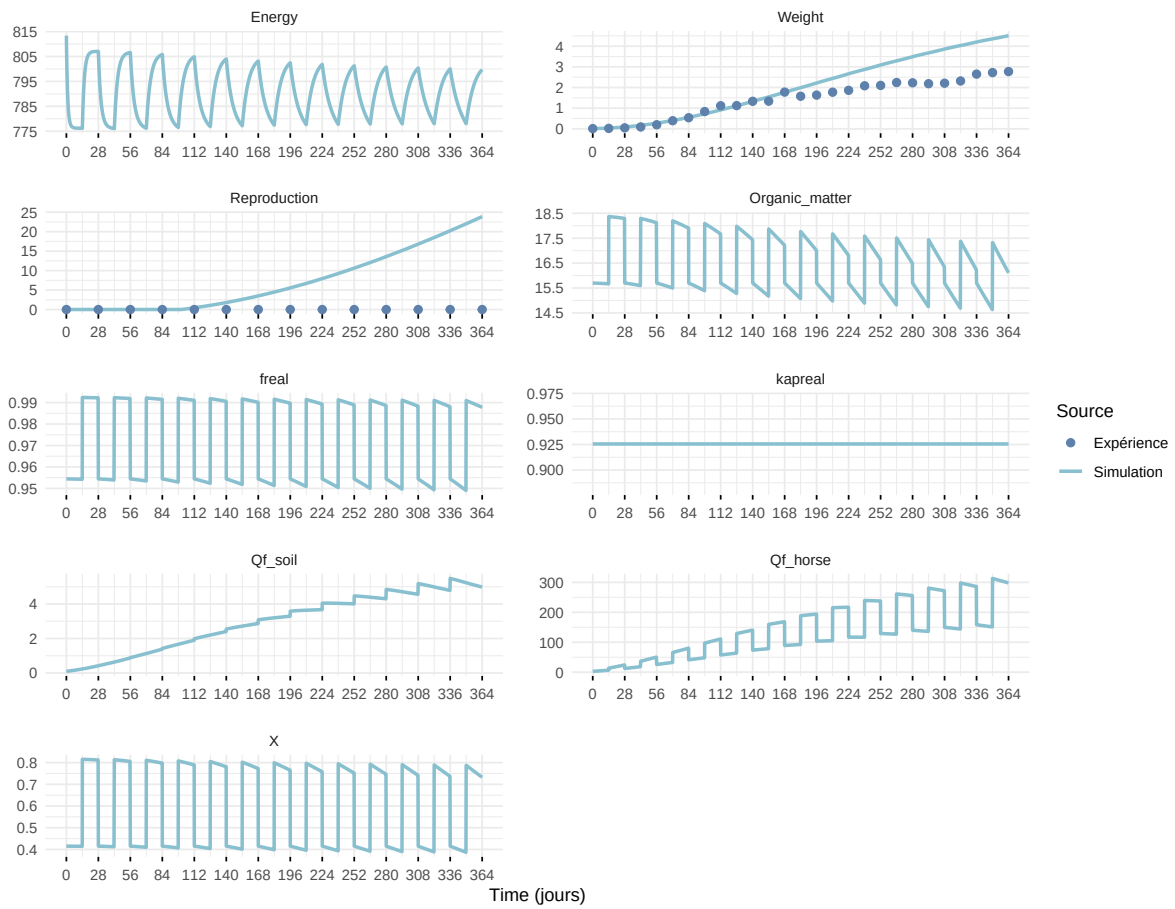
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexp
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship =

"many-to-many" to silence this warning.

Expérience : Bart2019_XP1_1



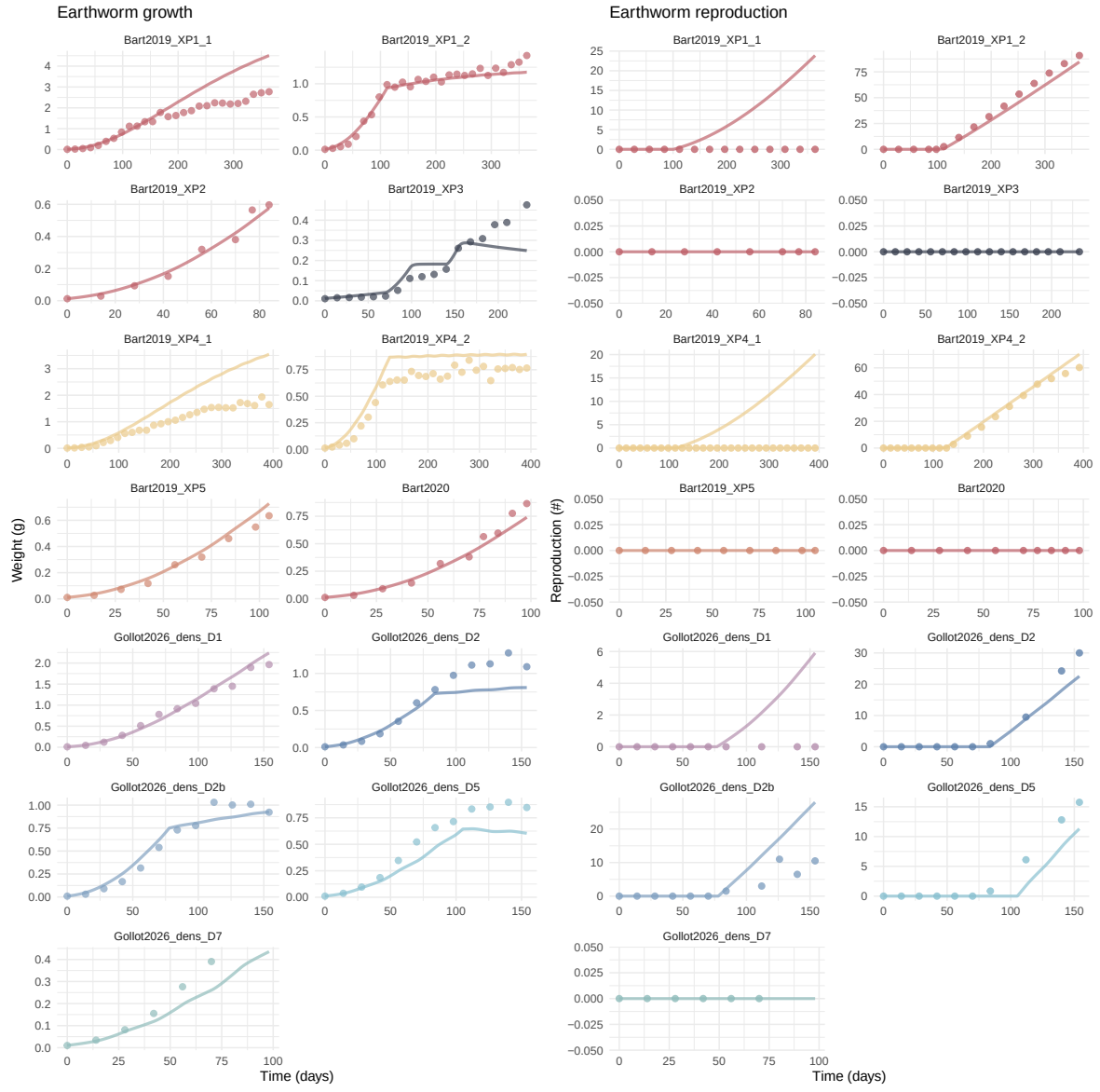
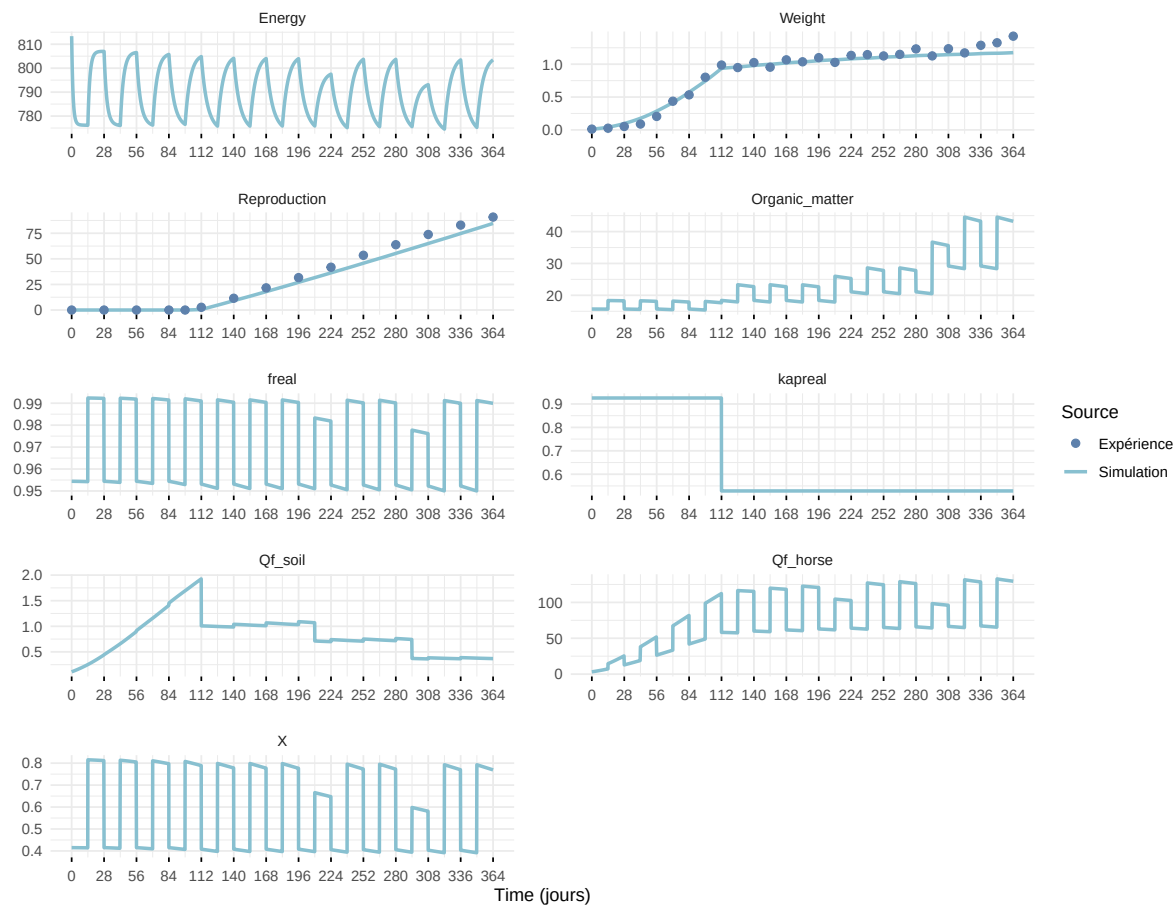
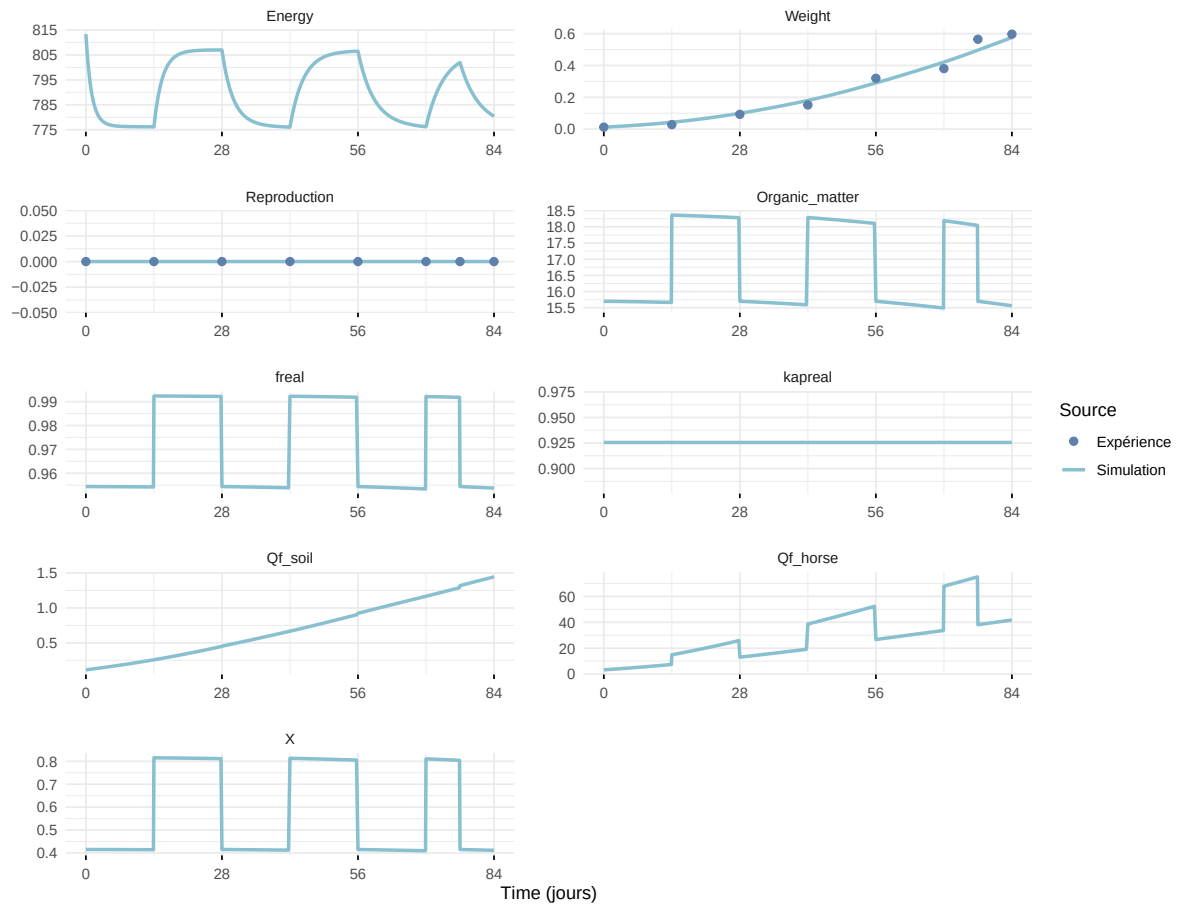


Figure 7: Simulations Weight and Reproduction.

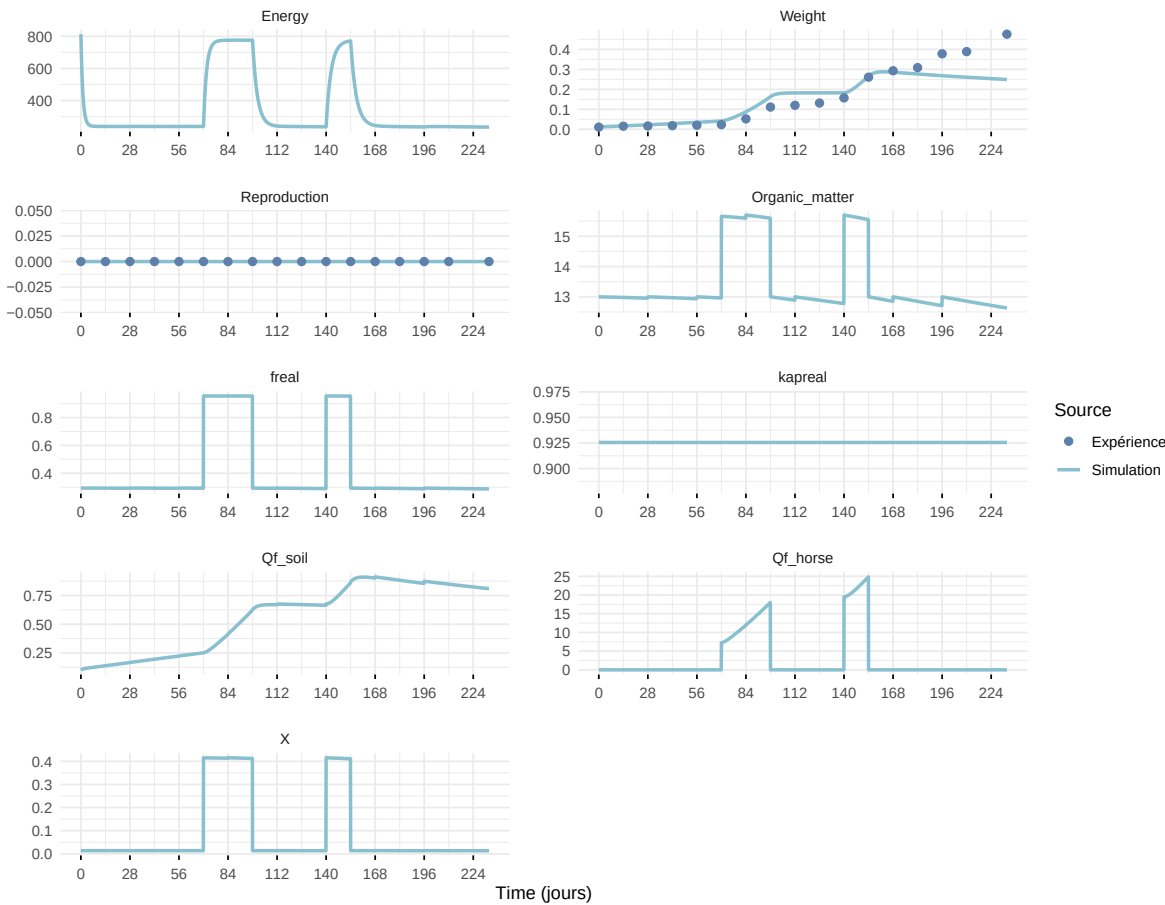
Expérience : Bart2019_XP1_2



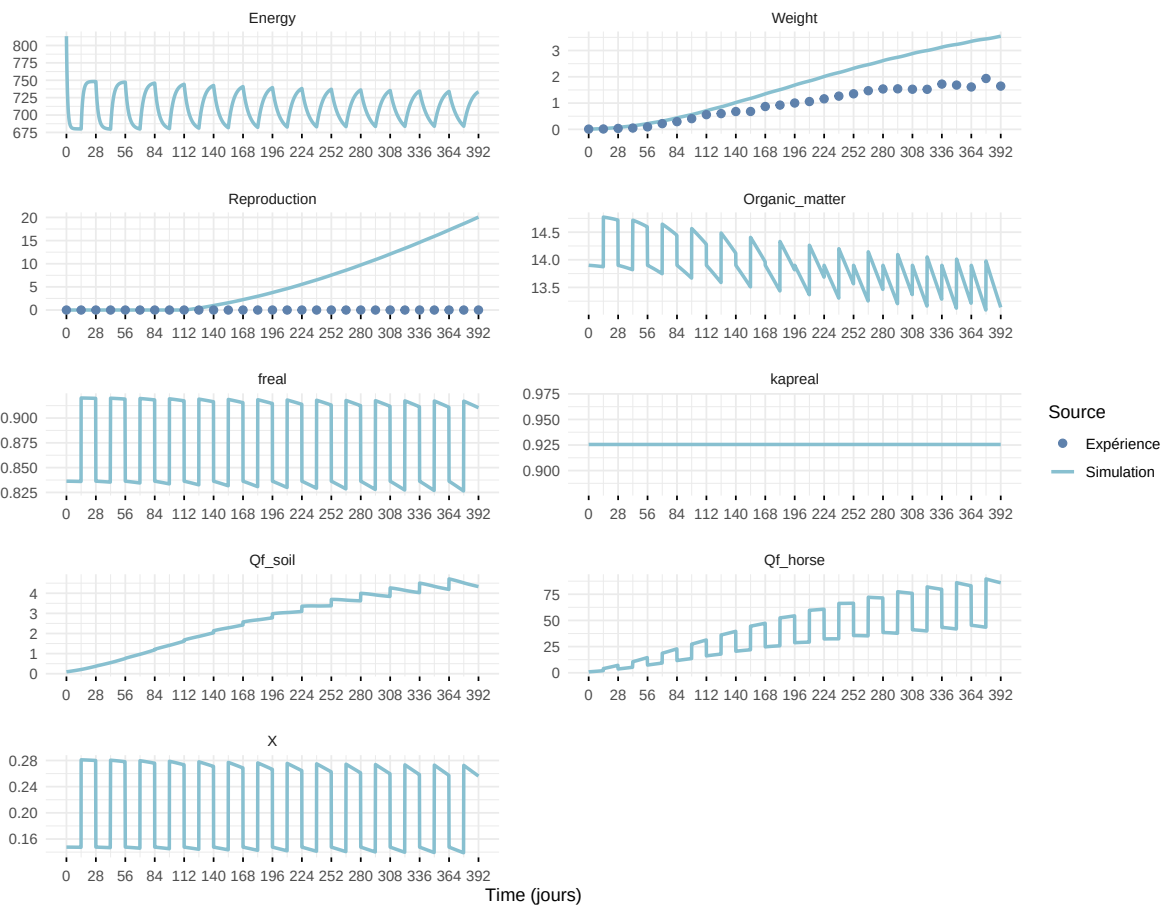
Expérience : Bart2019_XP2



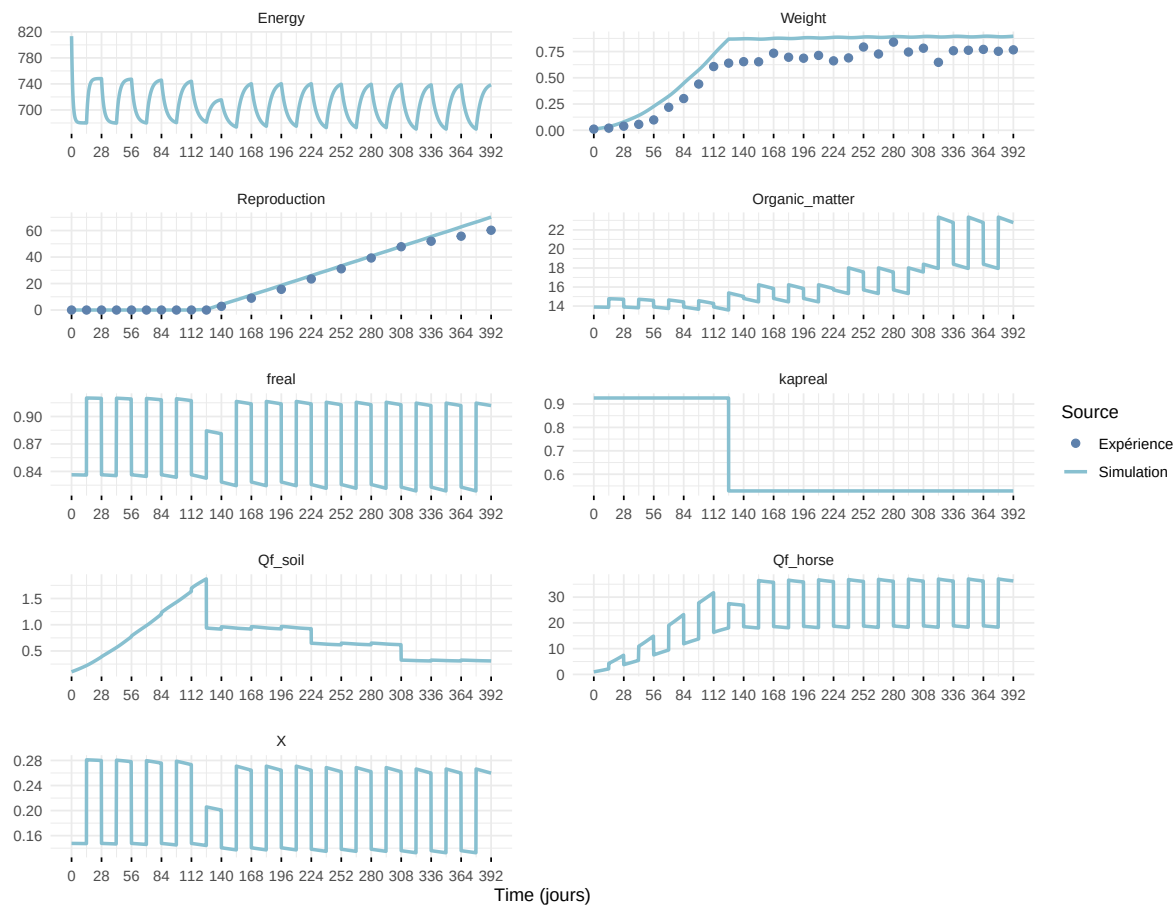
Expérience : Bart2019_XP3



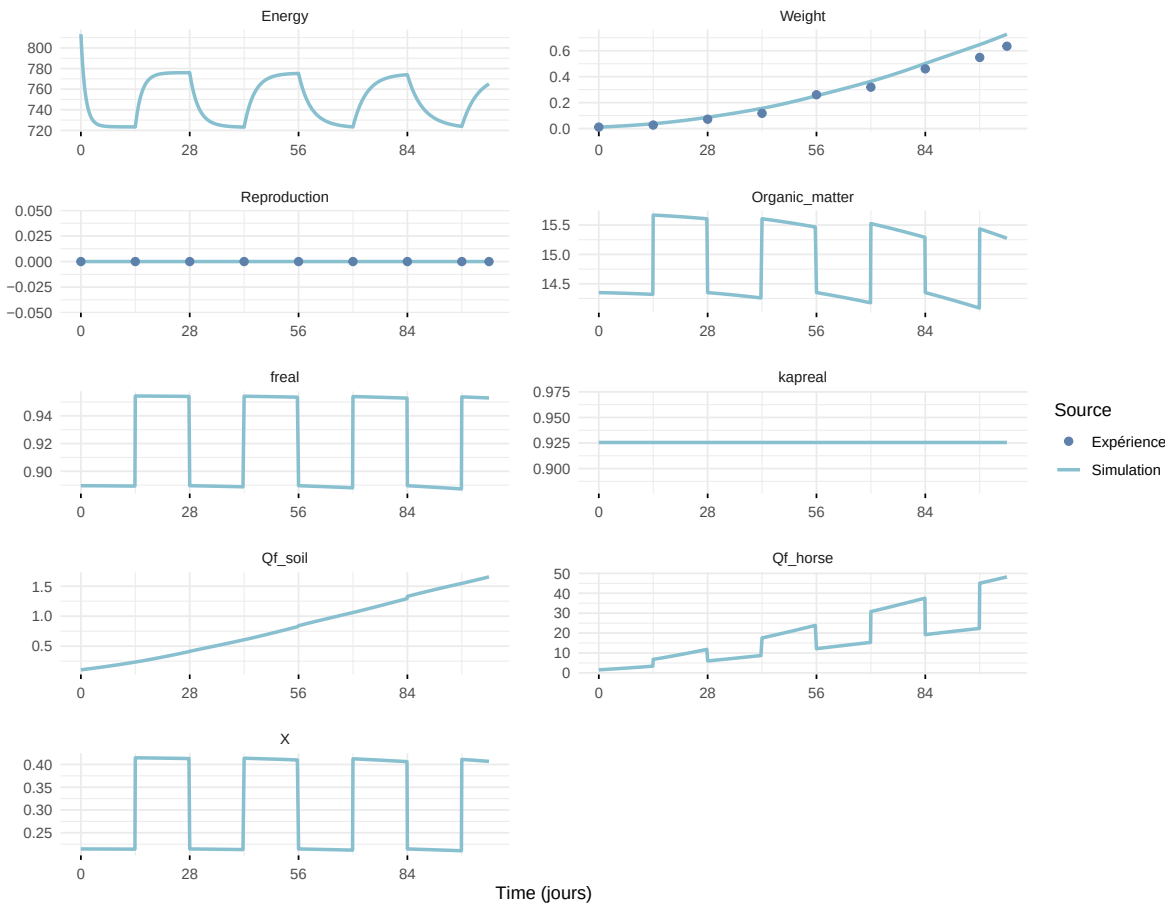
Expérience : Bart2019_XP4_1



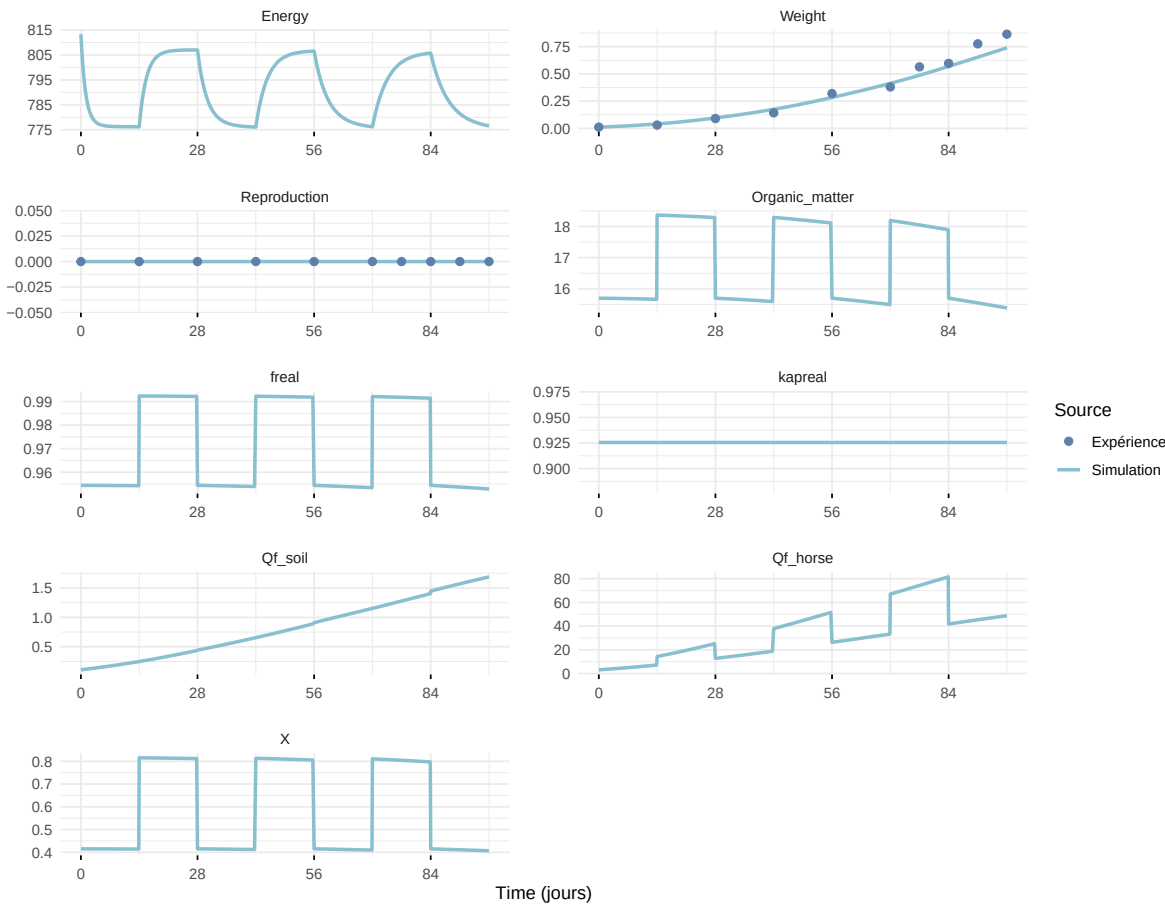
Expérience : Bart2019_XP4_2



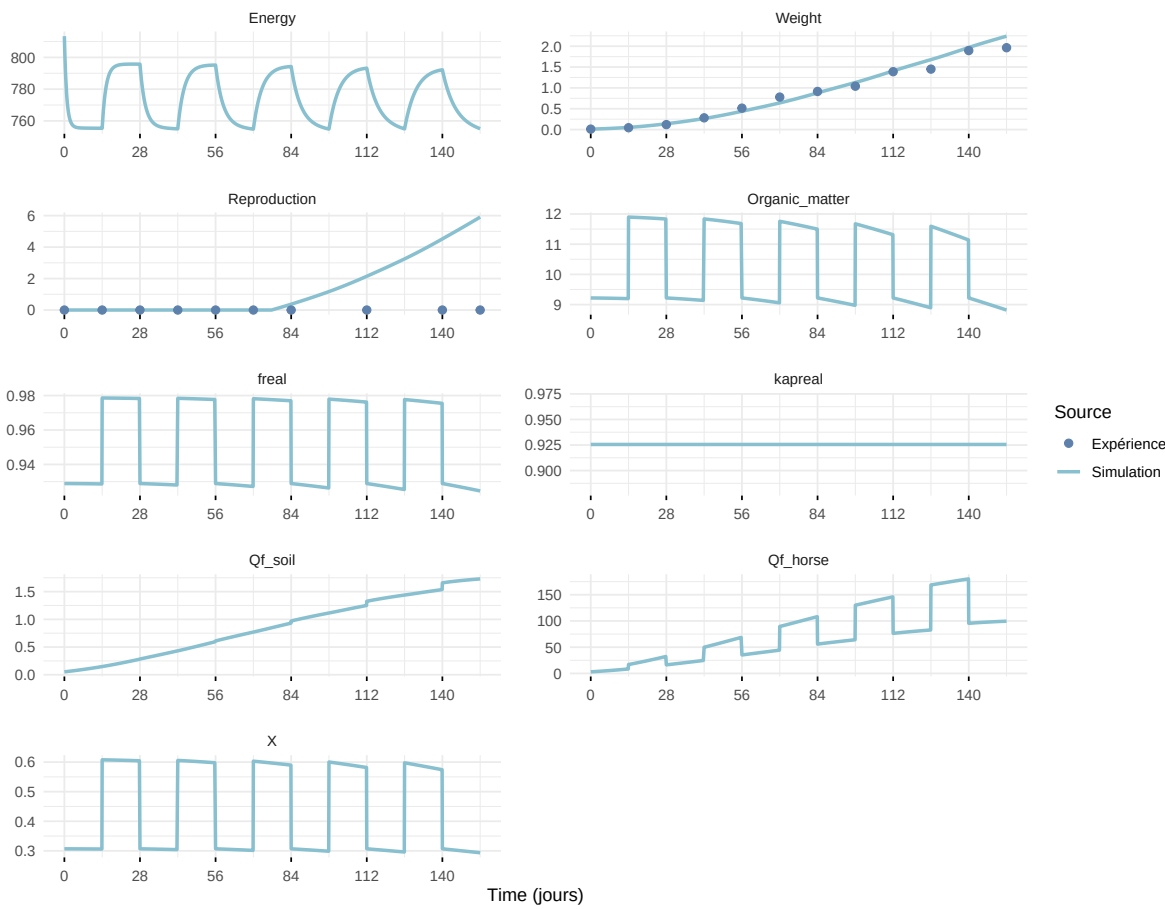
Expérience : Bart2019_XP5



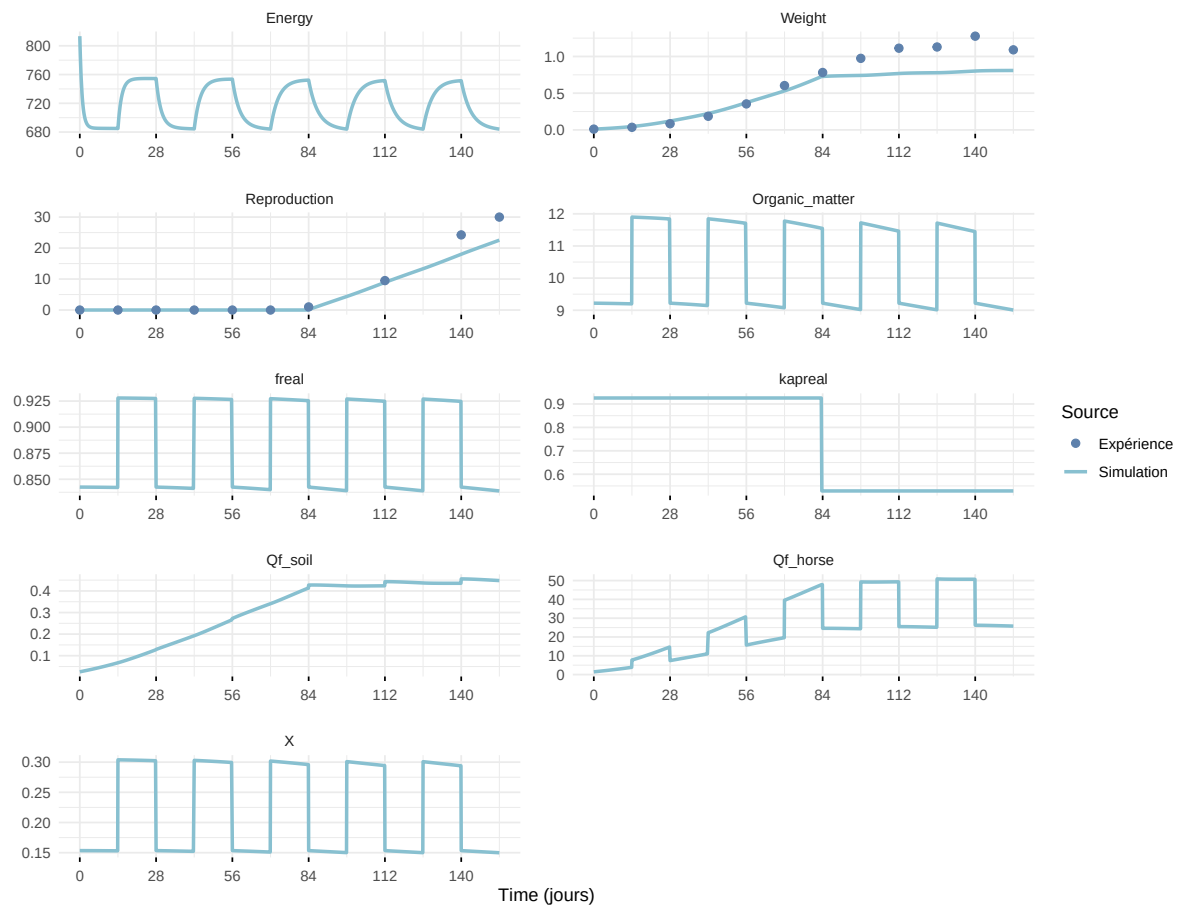
Expérience : Bart2020



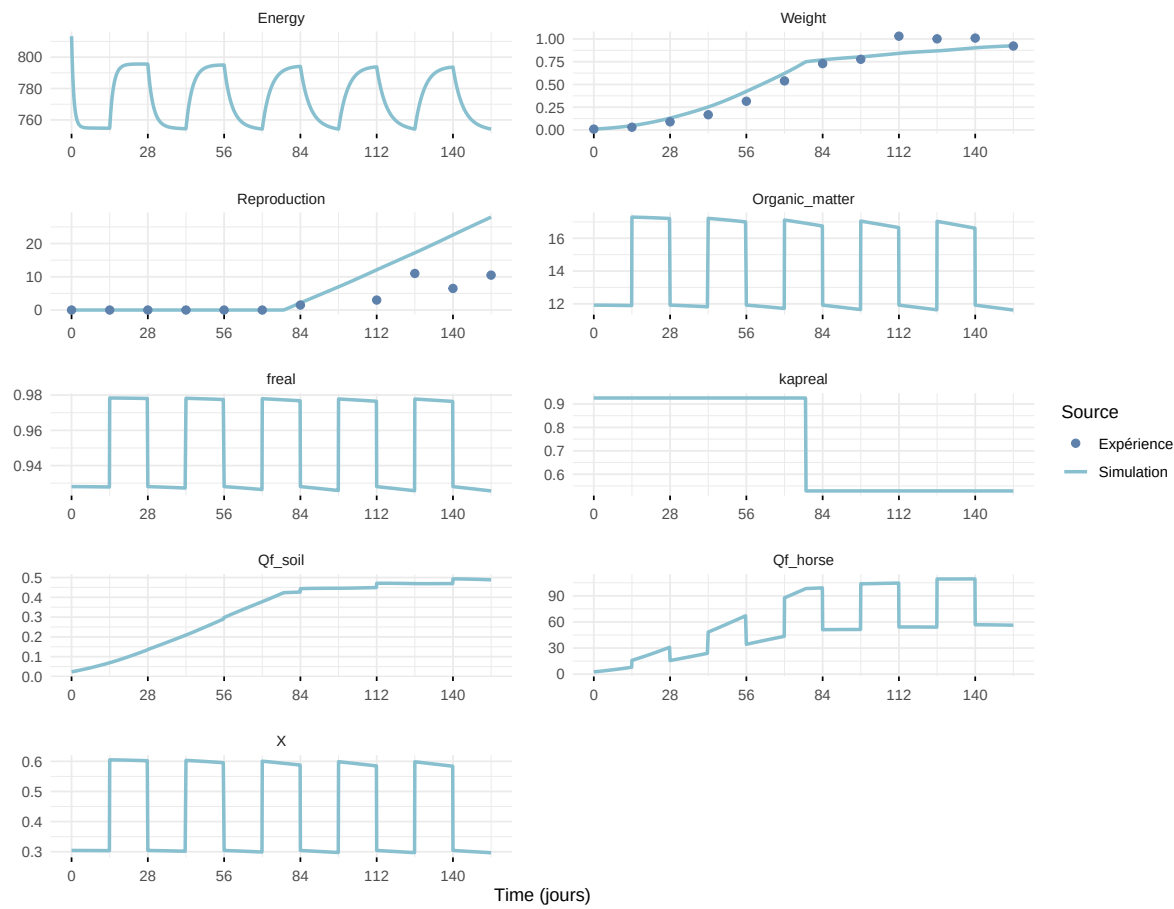
Expérience : Gollot2026_dens_D1



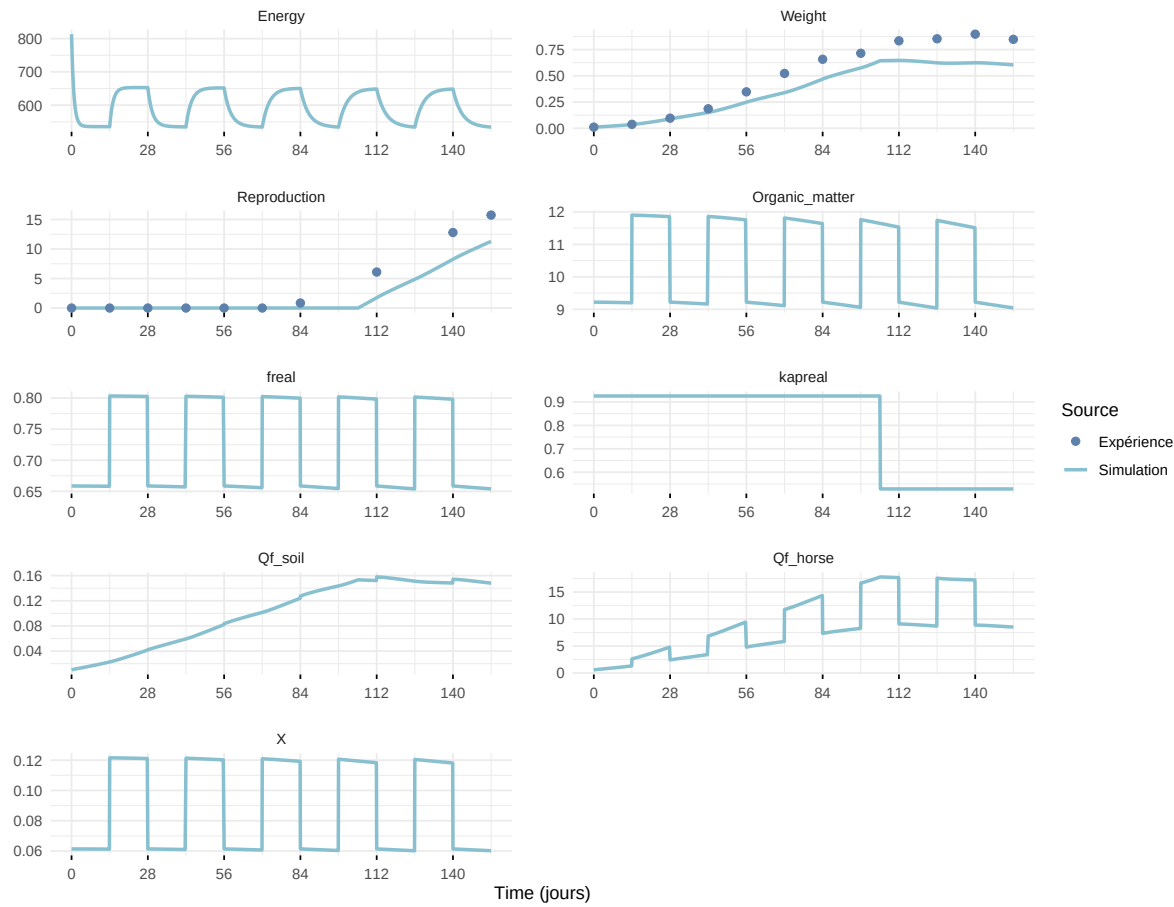
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

